

Algorithm Analysis And Data Structures

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Example for Running Time Calculation

Example:

$$\sum_{i=1}^N \sum_{j=i}^N \sum_{k=i}^j 1 = ?$$

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```

int sum(unsigned int n )
{
    int sum;

    sum = 0;
    for( int i=1; i<=n; i++ )
        for(int j=i; j<=n; j++ )
            for(int k = i; k<=j; k++ )
                sum += 1;

    return(sum );
}

```

Algorithm 1

$O(N^3)$

$$\sum_{k=i}^j 1 = j - i + 1$$

```
int sum(unsigned int n )
{
    int sum;

    sum = 0;
    for( int i=1; i<=n; i++ )
        for(int j=i; j<=n; j++ )
            sum += j-i+1;

    return(sum );
}
```

Algorithm 2

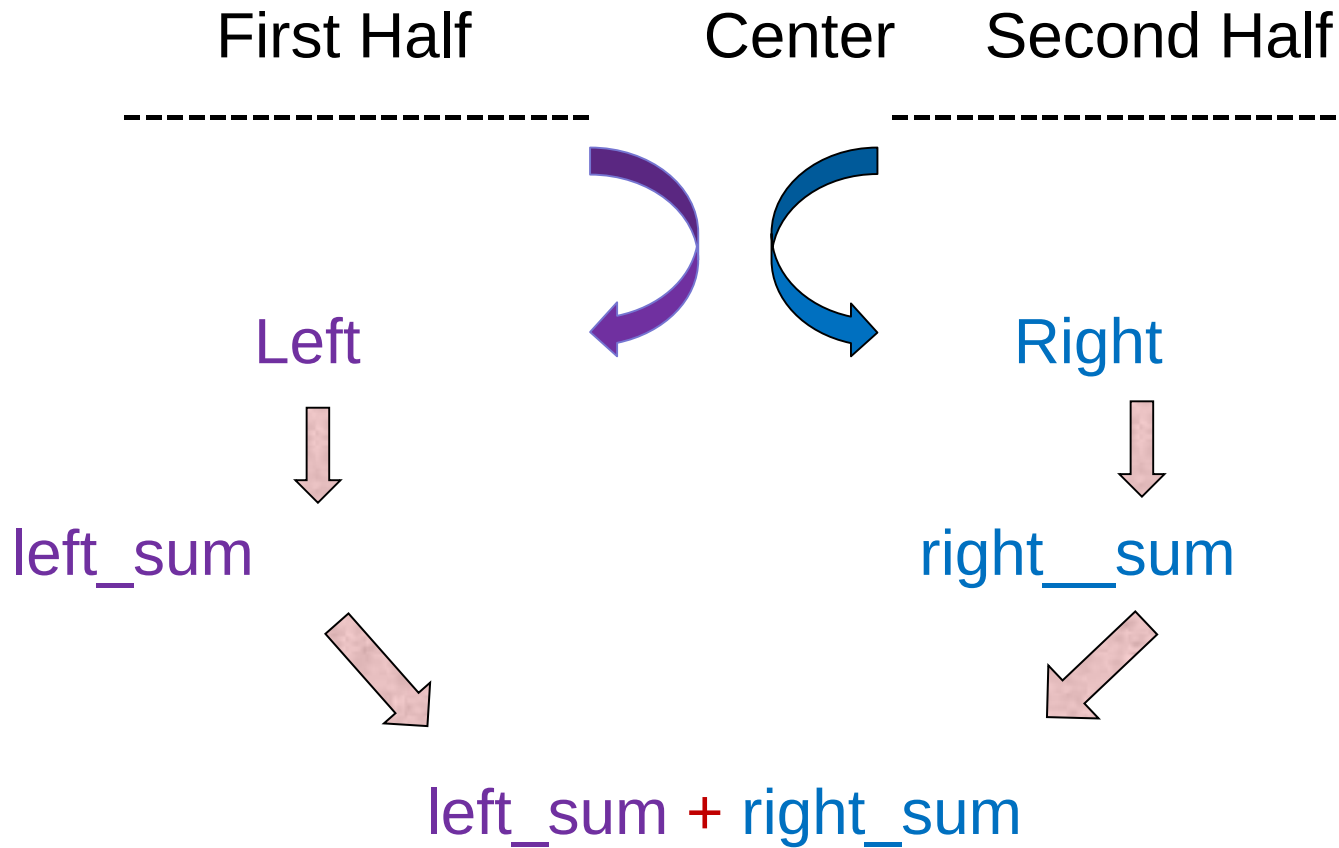
$O(N^2)$

$$\sum_{i=1}^N \sum_{j=i}^N \sum_{k=i}^j 1 = ?$$

$$\sum_{k=i}^j 1 = j - i + 1$$

$$\sum_{j=i}^N j - i + 1 = \sum_{p=1}^{N-i+1} p = \frac{(N-i+1).(N-i+2)}{2} \quad (p = j - i + 1)$$

Divide&Conquer Algorithm



```

int sum(int left, int right )
{
    int left_sum, right_sum;
    int center, sum;

    center = (left + right )/2;

    left_sum = 0;
    for(int i=center; i>=left; i-- )
        left_sum += (n-i+1)*(n-i+2)/2;

    center = center+1

    right_sum = 0;
    for(int i=center; i<=right; i++ )
        right_sum += (n-i+1)*(n-i+2)/2;

    sum = left_sum + right_sum;
    return(sum);
}

```

Algorithm 3 - Divide&Conquer Algorithm

$O(N\log N)$

$$\sum_{j=i}^N j - i + 1 = \sum_{p=1}^{N-i+1} p = \frac{(N-i+1).(N-i+2)}{2} \quad (p = j - i + 1)$$

```

int sum(unsigned int n )
{
    int sum;

    sum = 0;
    for( i=1; i<=n; i++ )
        sum += (n-i+1)*(n-i+2)/2;

    return(sum );
}

```

Algorithm 4

$O(N)$

$$\sum_{i=1}^N \sum_{j=i}^N \sum_{k=i}^j 1 = ?$$

$$\begin{aligned}
 & \sum_{i=1}^N \frac{(N-i+1).(N-i+2)}{2} \\
 &= \frac{1}{2} \sum_{i=1}^N i^2 - \left(N + \frac{3}{2} \right) \sum_{i=1}^N i + \frac{1}{2} (N^2 + 3N + 2) \sum_{i=1}^N 1 \\
 &= \frac{1}{2} \frac{N(N+1)(2N+1)}{6} - \left(N + \frac{3}{2} \right) \frac{N(N+1)}{2} + \frac{N^2 + 3N + 2}{2} N \\
 &= \frac{N^3 + 3N^2 + 2N}{6}
 \end{aligned}$$

```
int sum(unsigned int n )  
{  
    int sum;  
  
    sum = 0;  
    sum += (n*n*n + 3*n*n + 2*n) / 6;  
  
    return(sum );  
}
```

Algorithm 5

$O(12) = O(c)$

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Questions and Answers

Thank you

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